

Personalized Deep Learning Model for Closed-Loop 40 Hz Entrainment to Optimize Theta-Gamma Phase Amplitude Coupling in Alzheimer's Disease

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Introduction

Alzheimer's disease affects over 55 million people worldwide, with no cure and limited treatment options. Recent neuroscience research has revealed how 40 Hz auditory stimulation can synchronize brain gamma oscillations, reducing amyloid-beta plaques and tau tangles in animal models and showing early cognitive benefits in human clinical trials.

Current clinical protocols deliver this stimulation on a rigid fixed schedule — 40 seconds of 40 Hz auditory stimulation followed by 20 seconds of silence, cycling continuously for one hour regardless of how the patient's brain is responding. This one-size-fits-all approach ignores how some patients maintain strong theta-gamma coupling throughout a session, while others lose coupling within seconds and never fully synchronize. Prolonged repetitive stimulation causes neural habituation, where the brain progressively "tunes out" the stimulus, and coupling degrades over time. The result is wasted stimulation during periods of strong coupling and missed opportunities during periods of fading coupling.

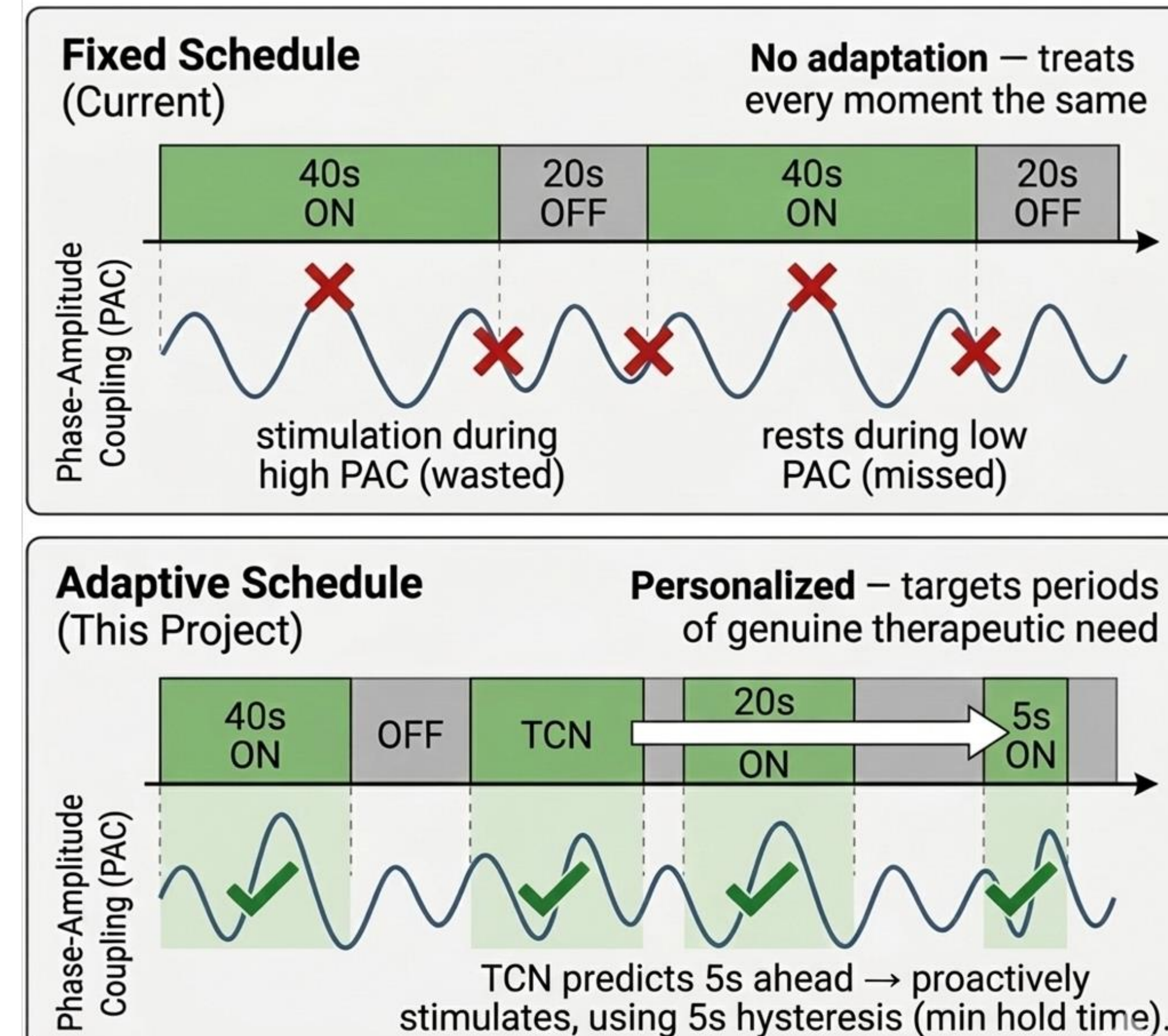
This project asks: can we predict when a patient's brain will lose entrainment and deliver stimulation proactively, before the decline occurs?

Background

- 40 Hz gamma entrainment reduces amyloid load in Alzheimer's mouse models by activating microglia (Iaccarino 2016); multi-sensory stimulation reduces tau pathology and improves cognition (Martorell 2019); auditory entrainment enhances default mode network connectivity in human dementia patients (Lahijanian 2024)
- Phase-Amplitude Coupling (PAC)** quantifies entrainment success — how strongly gamma amplitude (38-42 Hz) locks to theta phase (4-8 Hz), measured via the Modulation Index (Tort 2010). High PAC = entrained; low PAC = lost synchronization.
- Neural habituation causes repeated identical stimuli to produce progressively weaker neural responses (Thompson & Spencer 1966), making fixed schedules less effective over time

No existing system predicts **future entrainment** state for proactive control. Current approaches either use fixed schedules (no adaptation) or reactive thresholds (respond after PAC decline has already occurred).

FIGURE 1: Fixed vs. Adaptive Scheduling



Hypothesis

A causal temporal convolutional network can predict future PAC at 5-10 second horizons where simpler methods fail, enabling adaptive stimulation that targets periods of genuine therapeutic need.

Goals:

- Predict brain entrainment 5-10s into the future (the minimum lead time for proactive control)
- Closed control loop pipeline — integrate predictions into a closed-loop stimulation controller
- Validate on real patient EEG that predictive control outperforms fixed and reactive approaches

References

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- Lawhern, V. J., et al. (2018). EEGNet: A compact convolutional neural network for EEG-based brain-computer interfaces. *Journal of Neural Engineering*, 15(5), 056013.
- Lahijanian, M., et al. (2024). Auditory gamma-band entrainment enhances default mode network connectivity in dementia patients. *Scientific Reports*, 14, 13153.
- Thompson, R. F., & Spencer, W. A. (1966). Habituation: A model phenomenon for the study of neuronal substrates of behavior. *Psychological Review*, 73(1), 16-43.

Acknowledgements

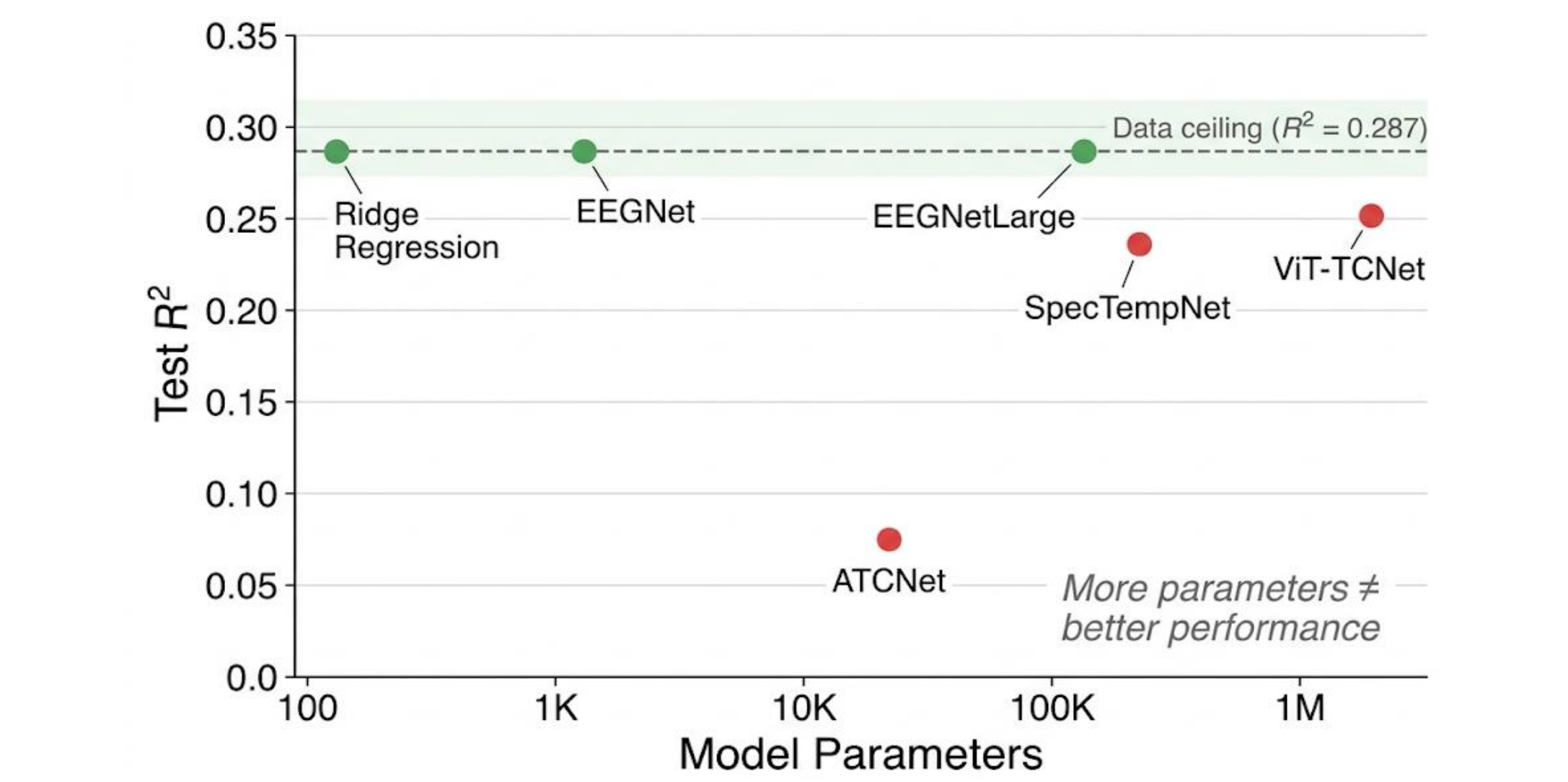
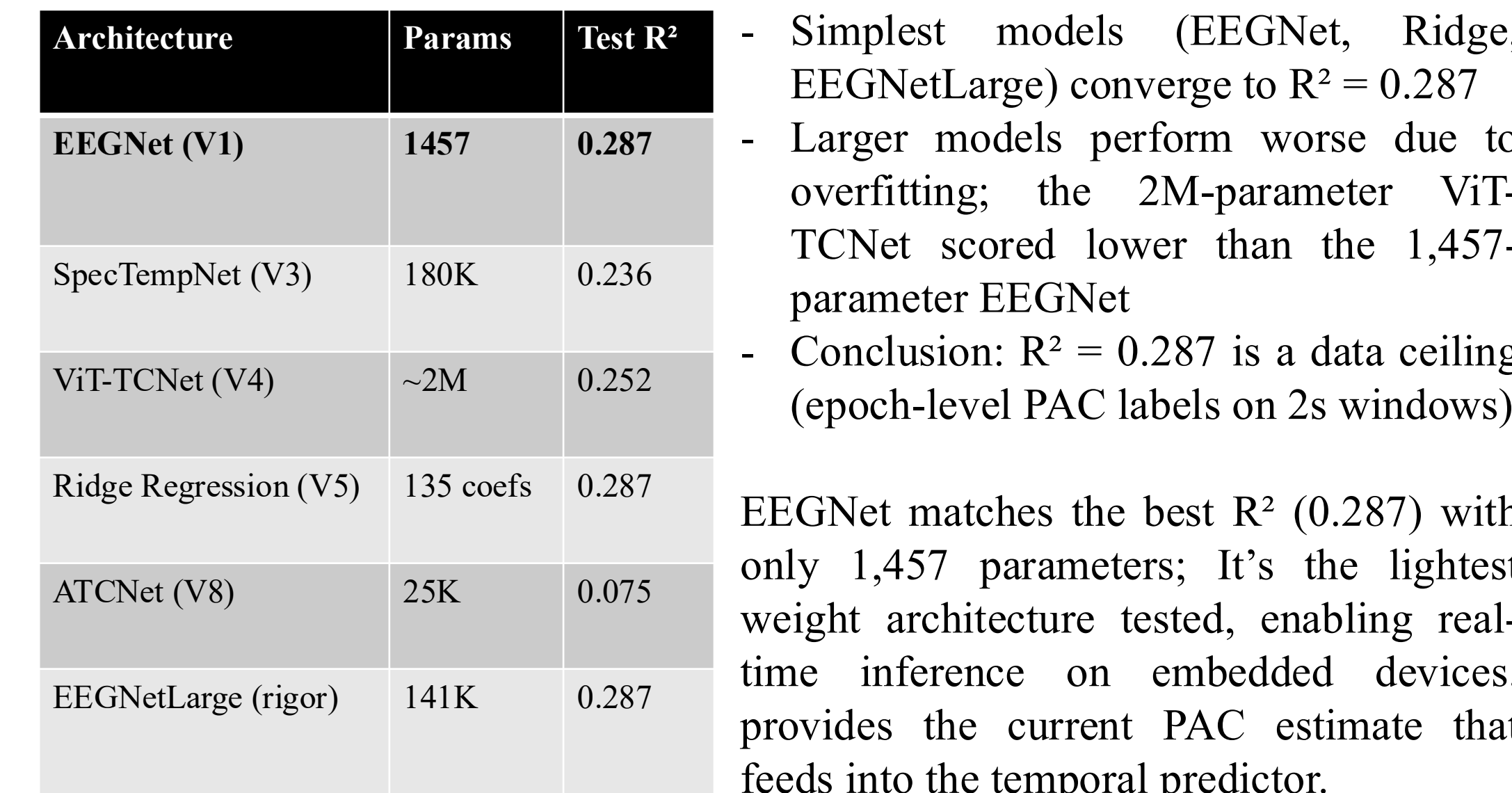
Data Source: OpenNeuro dataset ds005048 (Lahijanian et al., 2024). All EEG data used under open access license.

Model Approach

To control stimulation proactively, the system needs to predict PAC 5-10 seconds into the future. The approach consists of two phases:

- Estimate current PAC from raw EEG in real time (since live patients don't have ground-truth labels)
- Predict future PAC from a history of past estimates — temporal forecasting

Stage 1: Architecture Decision for Static PAC Estimation

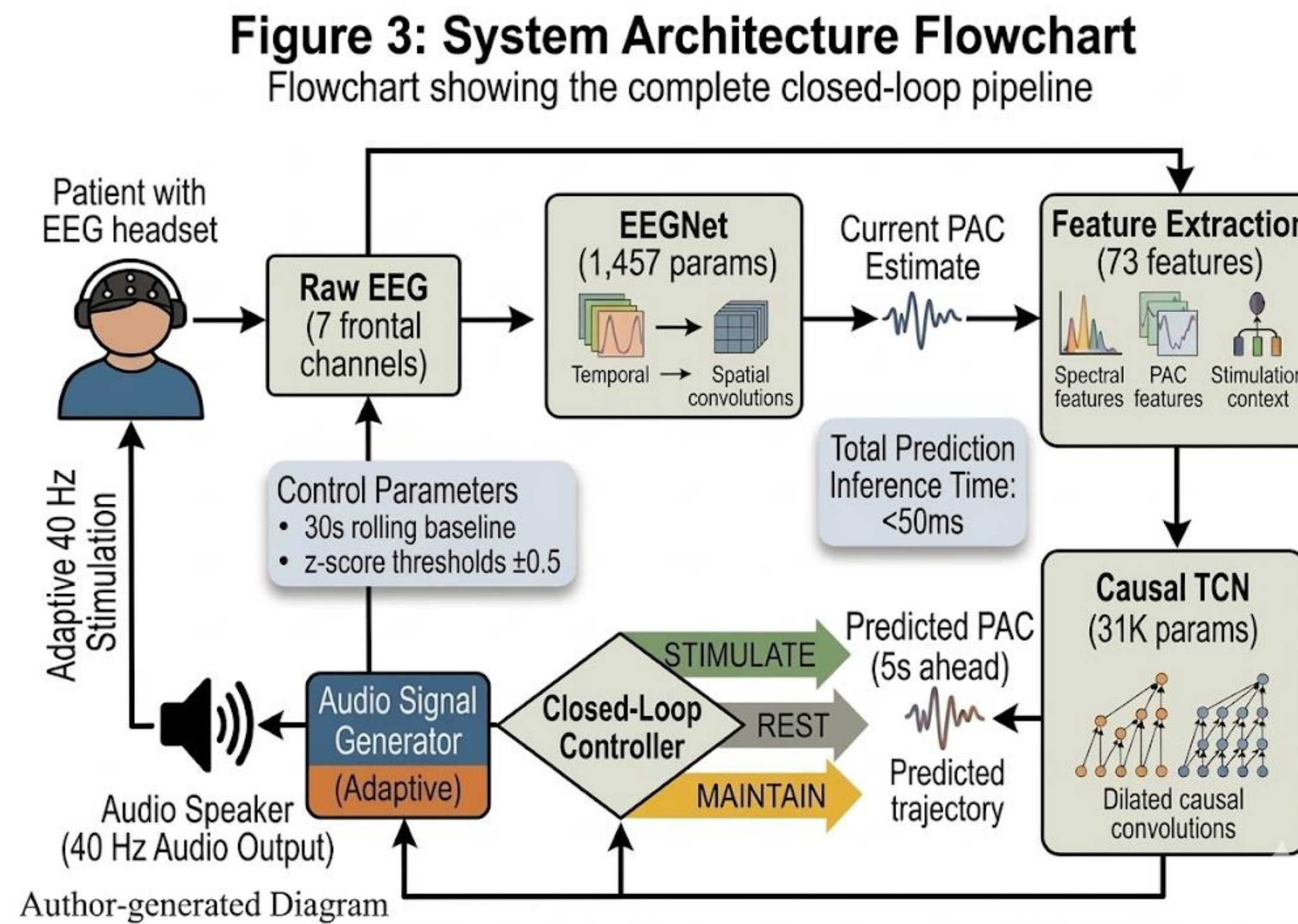


Stage 2: Temporal Prediction; Causal TCN

Since no single-window architecture could exceed $R^2 = 0.287$, I shifted the approach: instead of predicting PAC better from one snapshot, predict it further into the future from a sequence of snapshots. The Causal TCN ingests 20 seconds of history and forecasts PAC 5 seconds ahead.

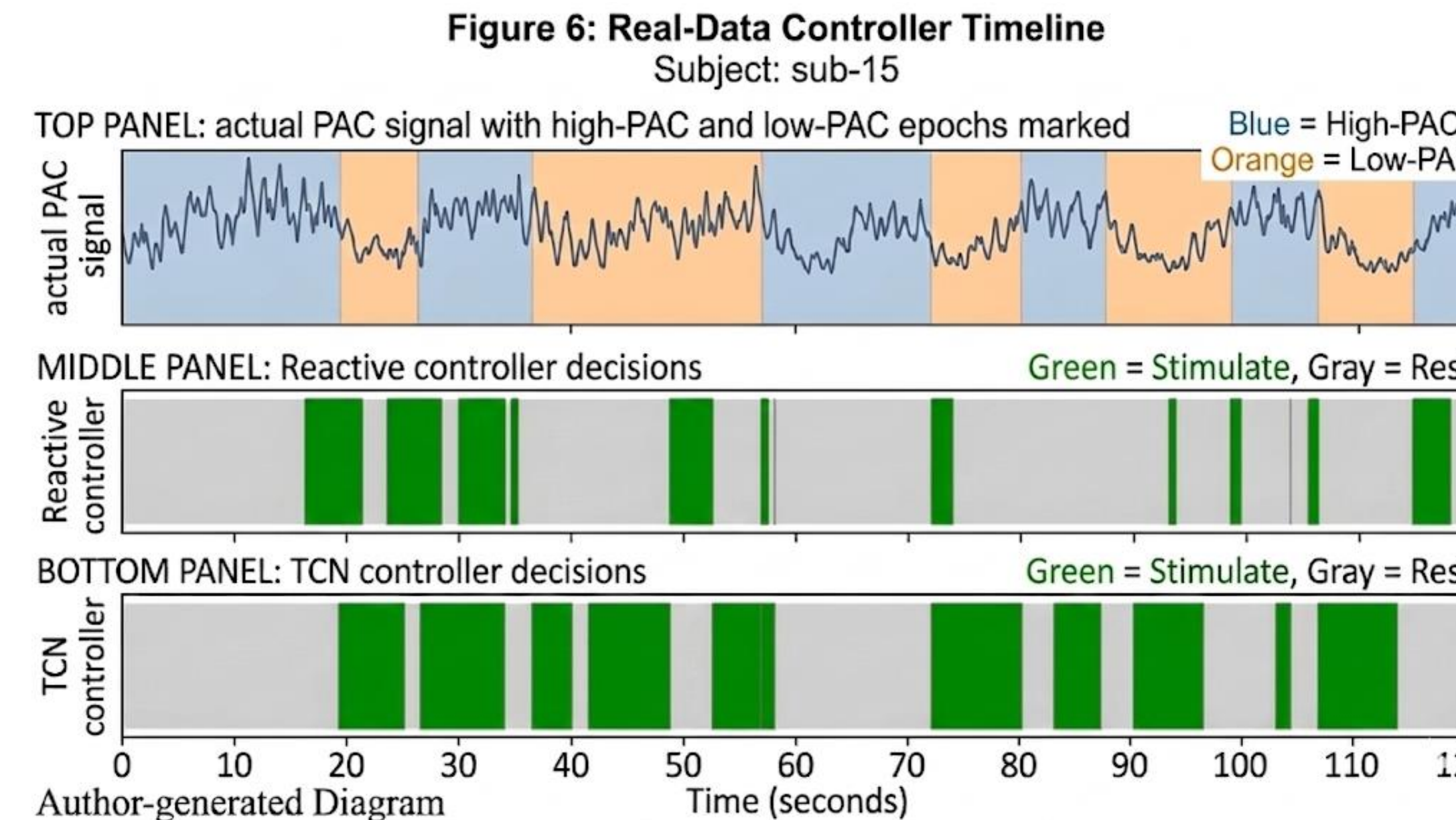
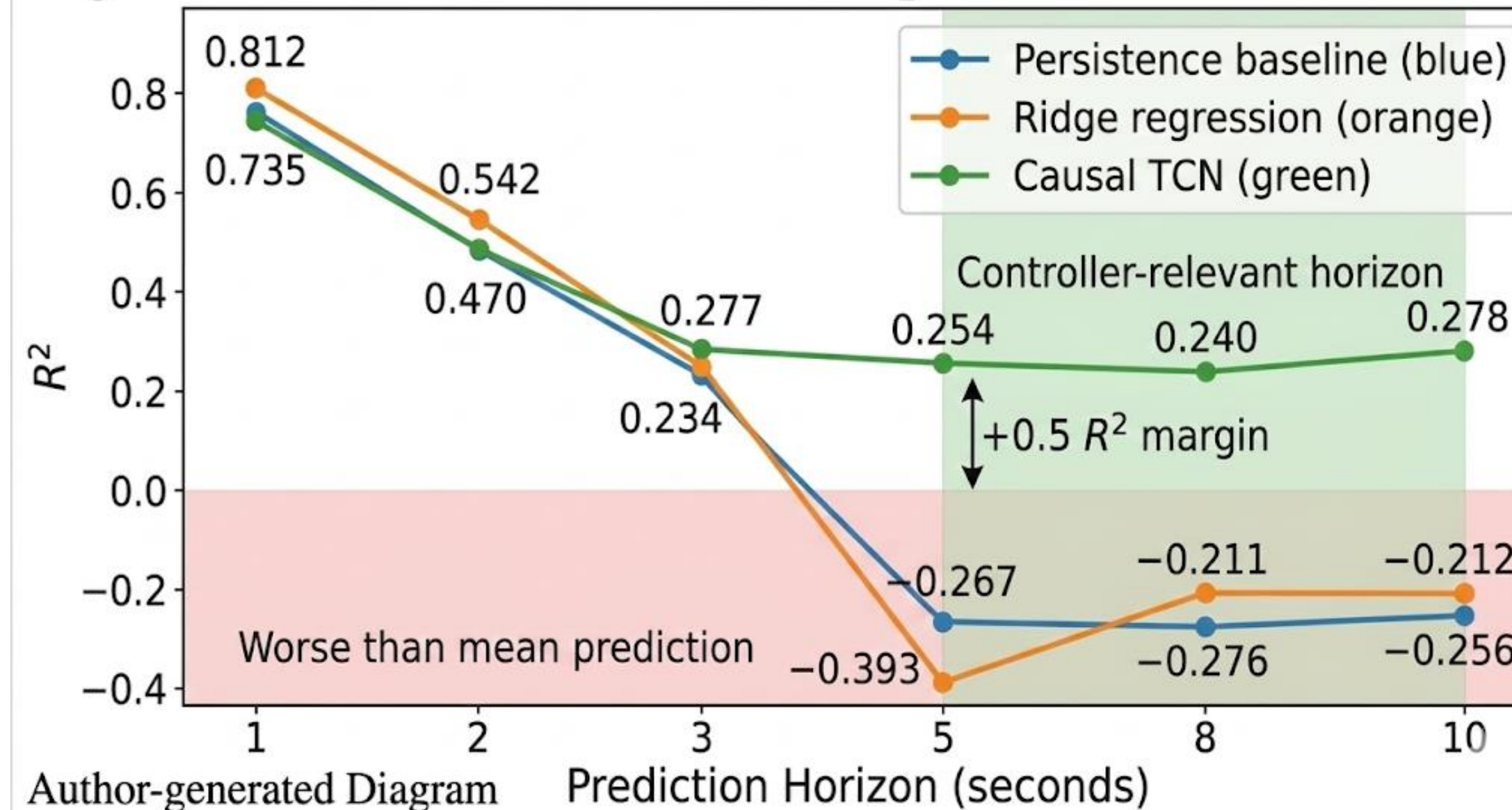
	EEGNet	Causal TCN
Purpose	Estimate current PAC from raw EEG	Predict future PAC from history
Parameters	1457 (12 samples/param)	31,043
Input	Raw EEG (7 ch x 500 samples)	73 features x 20 timesteps
Architecture	Temporal + depth wise spatial conv.	Dilated causal conv (d=1,2,4,8)
Output	Current PAC Estimate	Future PAC (5s Horizon)
Key Feature	Real-time-inference	31-step receptive field, causal (no future leakage)

73 TCN input features: 61 spectral power + 7 PAC-derived (current value from EEGNet, moving averages, trends) + 5 stimulation context (on/off state, duration, session phase)



Data, Charts, & Models

Figure 5: Prediction Horizon Sweep

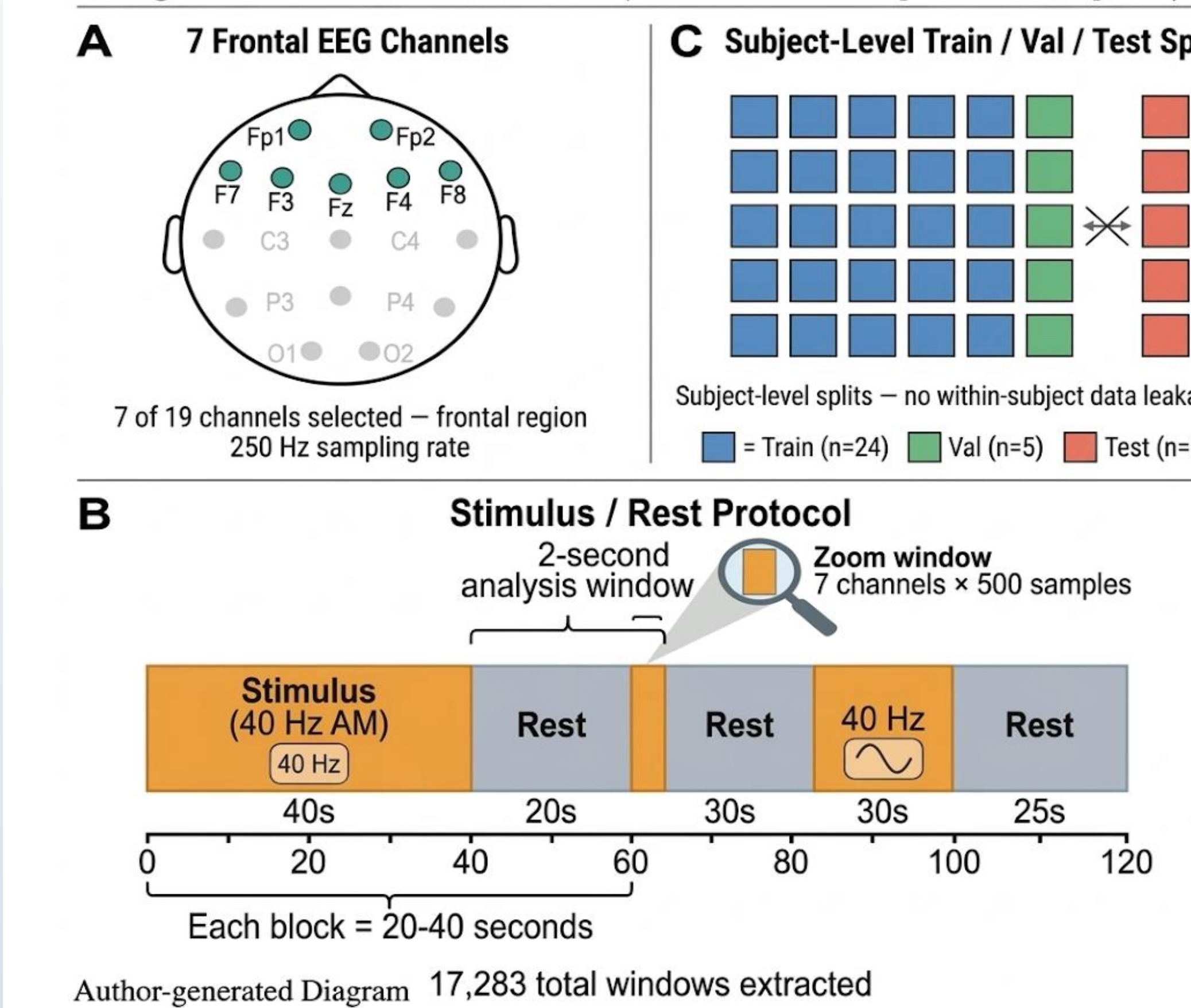


Materials

- Dataset:
- 35 dementia patients, OpenNeuro ds005048 (Lahijanian 2024)
 - 7 frontal EEG channels (Fp1, Fp2, F3, F4, F7, F8, Fz), 250 Hz
 - Alternating Stimulus (40 Hz AM auditory) and Rest epochs, 20-40s each
 - 17,283 two-second windows; subject-level splits (24 train / 5 val / 6 test)

Compute: Python 3.13, PyTorch, Apple Silicon (MPS), no cloud GPU required

Figure 2: Dataset Overview (EEG channels, protocol, splits)



Procedures

Training:

- EEGNet: MSE loss, Adam optimizer, gradient clipping (max_norm=1.0), early stopping
- Causal TCN: Huber loss (robust to PAC outliers), learning rate reduction on plateau, early stopping (patience=20)
- All hyperparameters tuned on validation set (5 subjects); test set (6 subjects) used only for final evaluation

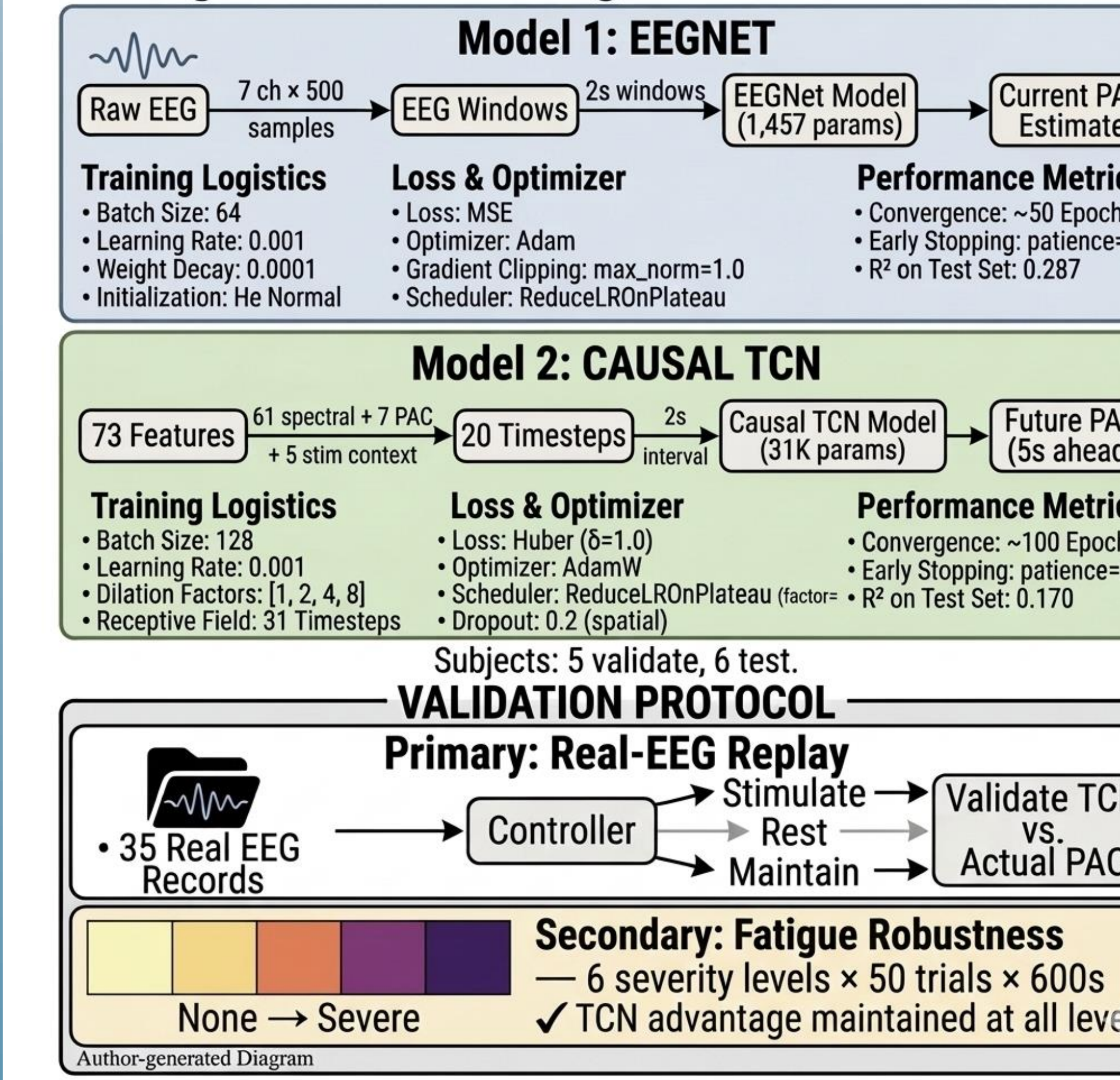
Validation (two independent protocols):

- Primary (real data): Replayed TCN controller on all 35 subjects' actual EEG recordings — offline counterfactual decisions on real brain data. For validation, ground-truth PAC labels were used as TCN input, isolating the TCN's predictive contribution from EEGNet estimation error.
- Secondary (simulation): Closed-loop simulation with neural fatigue modeling - 6 severity levels x 50 trials x 600 seconds per trial

Statistical validation:

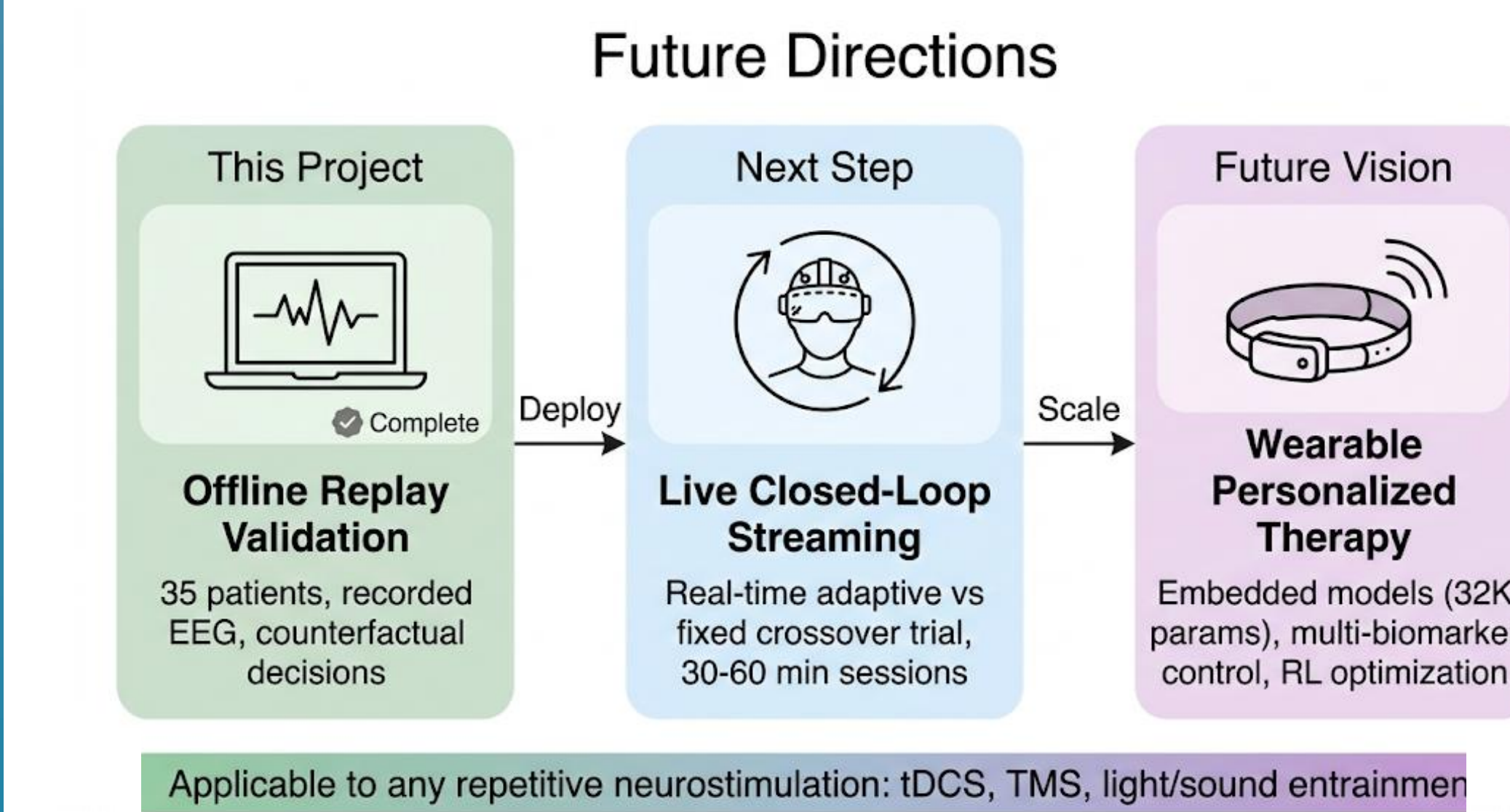
- Wilcoxon signed-rank tests (non-parametric, paired)
- Bootstrap 95% confidence intervals
- Hedges' g effect sizes
- Compared against: Fixed Schedule, Reactive Threshold, PI Controller, Alignment Oracle (theoretical upper bound)

Figure 4: Full Training & Validation Protocol



Further Research

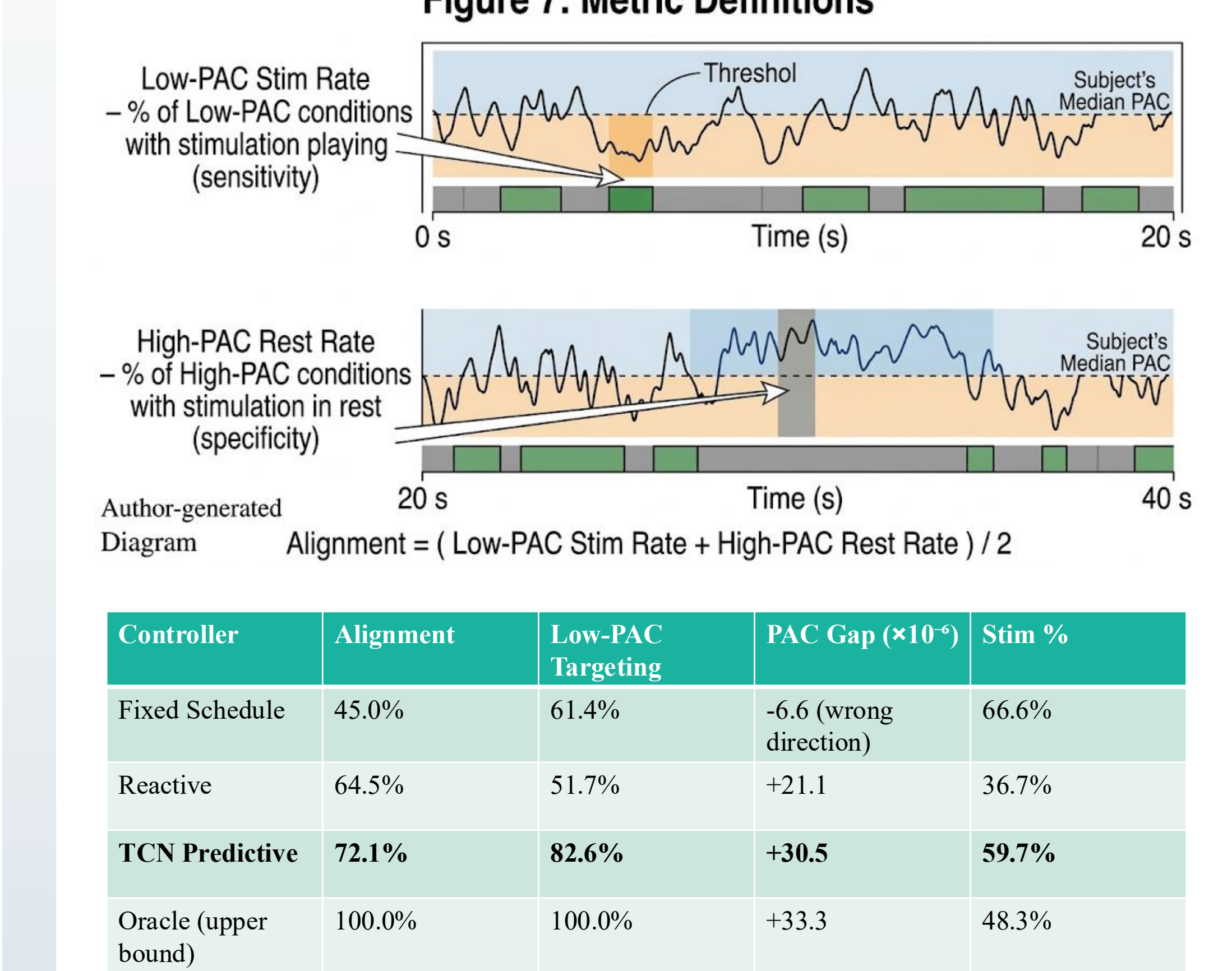
- Deploy with live EEG streaming for real-time crossover validation (adaptive vs fixed within same session)
- Record 30-60-minute sessions to capture the full habituation time course
- Replace the heuristic controller with reinforcement learning for long-horizon optimization
- Extend to multi-biomarker control (PAC + spectral power + connectivity)



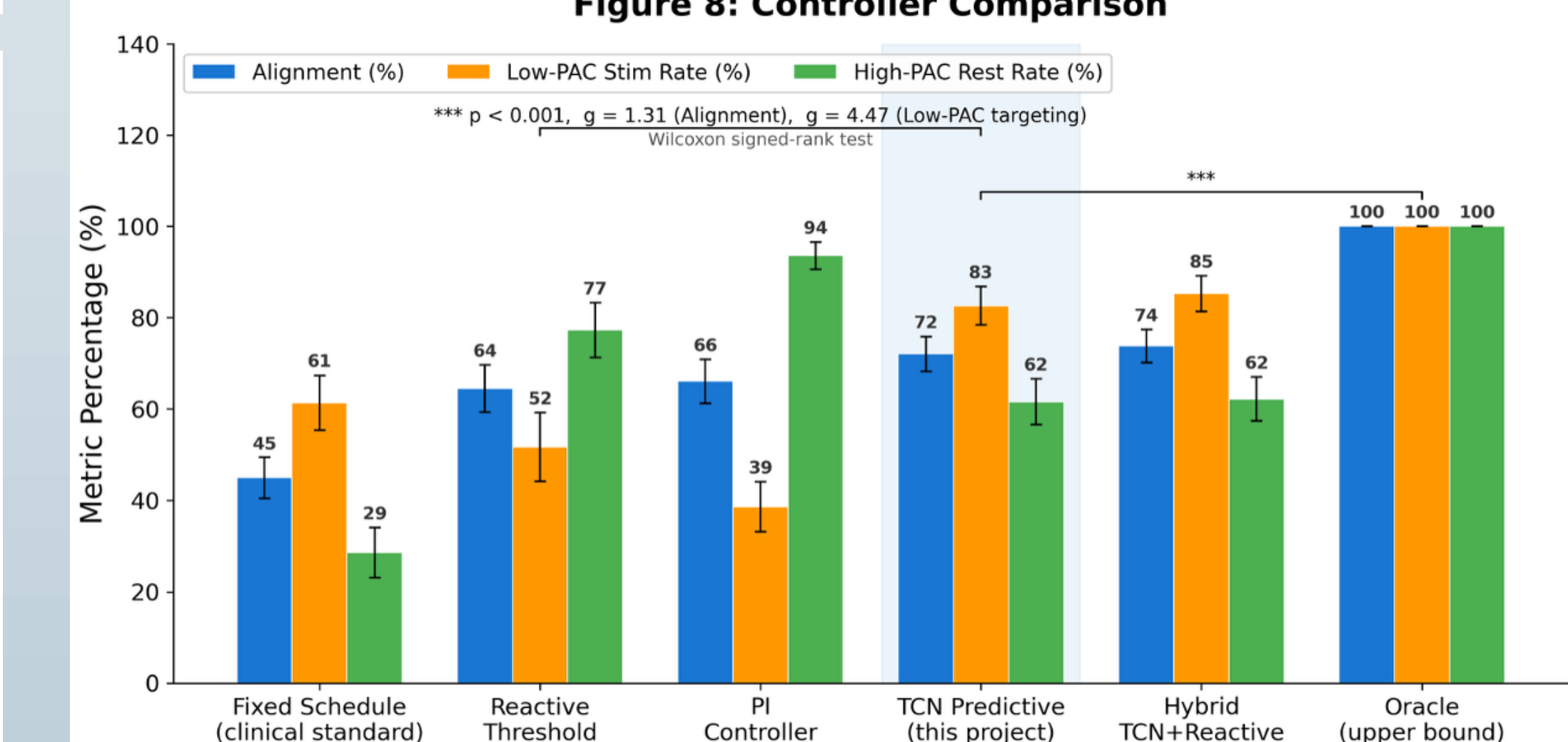
Results & Findings

Result 1: TCN Predictive Control Outperforms All Alternatives

Controller performance replayed on all 35 subjects' real EEG:

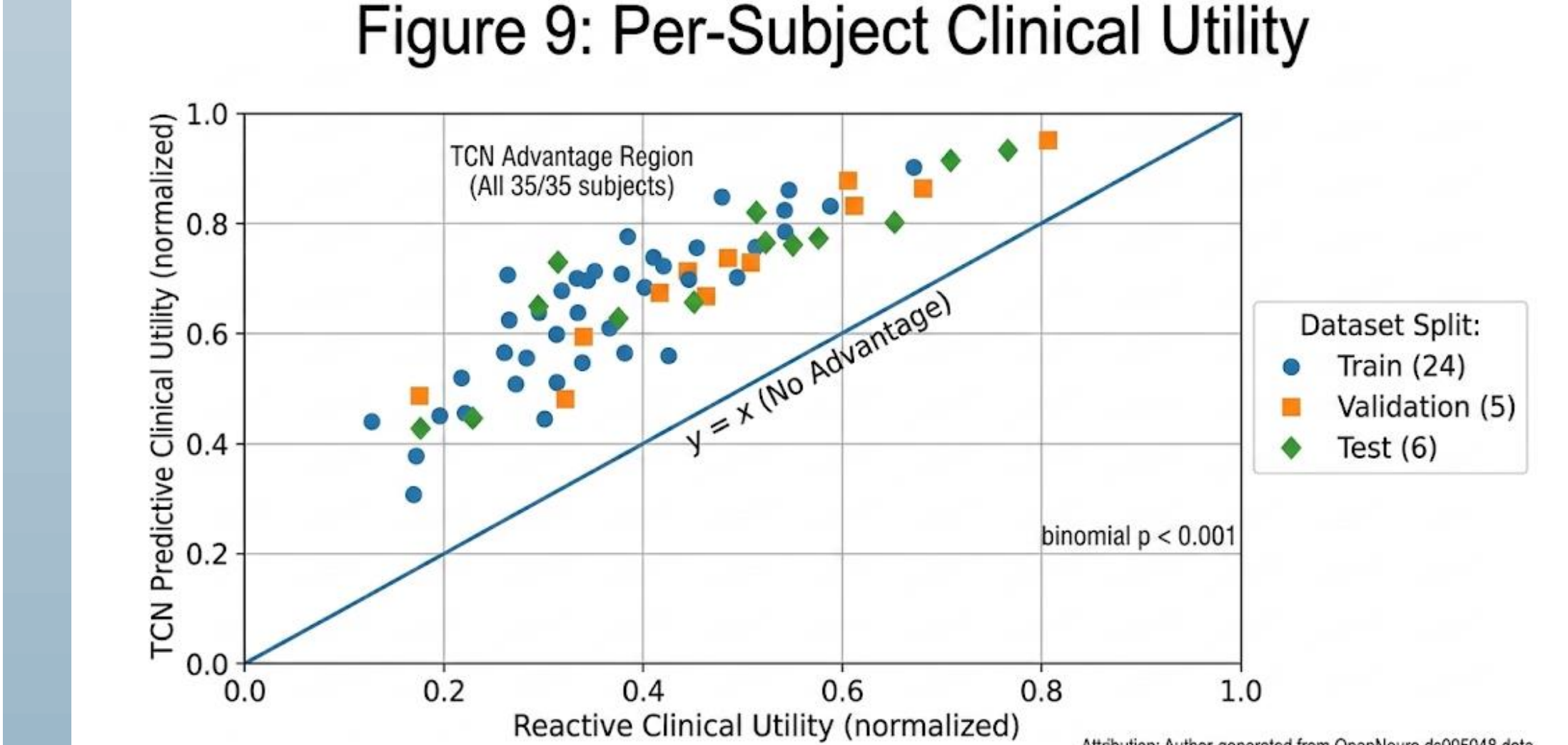


Controller	Alignment	Low-PAC Targeting	PAC Gap (x10 ⁻¹)	Stim %
Fixed Schedule	45.0%	61.4%	-6.6 (wrong direction)	66.6%
Reactive	64.5%	51.7%	+21.1	36.7%
TCN Predictive	72.1%	82.6%	+30.5	59.7%
Oracle (upper bound)	100.0%	100.0%	+33.3	48.3%



Result 2: Every Patient Benefits

Figure 9: Per-Subject Clinical Utility



- 35/35 subjects (100%) show higher utility with TCN vs Reactive (binomial $p < 0.001$)
- Advantage holds for 6 held-out test subjects never seen during training

Result 3: Advantage Increases with Fatigue

Simulation results across 6 fatigue severity levels (n = 50 trials each):

Fatigue Level	Fixed Efficiency	Adaptive Efficiency	Improvement
None	0.343	0.375	+9.5%***
Mild	0.341	0.375	+10.0%***
Moderate	0.335	0.366	+9.0%***
High	0.319	0.354	+10.8%***
Severe	0.316	0.352	+11.2%***

*** $p < 0.001$, Hedges' $g = 1.7-2.4$ (large to very large effects). Advantage grows as fatigue worsens; exactly when personalization matters most.

Result 4: Robust Across Fatigue Model Assumptions

Tested under 4 different mathematical models of neural fatigue (n = 50 trials each):

Fatigue Model	Advantage	P-value	Hedges' g
Exponential Decay	+9.0%	1.8×10^{-15}	2.31
Step Function	+6.9%	4.4×10^{-14}	1.21
Heterogeneous (50/50)	+8.9%	2.5×10^{-14}	1.71
Saturation (synaptic)	+19.0%	1.8×10^{-15}	3.66

Conclusions

- At 5-10s horizons, the TCN maintains $R^2 \approx 0.25$ while all baselines collapse below zero; a +0.5 R^2 margin at the operationally relevant range for proactive control
- On real patient EEG: 72.1% alignment vs 64.5% reactive ($p < 0.001$), targeting 82.6% of low-PAC windows vs 51.7% = 60% improvement in therapeutic targeting
- All 35 patients benefited, including 6 held-out test subjects — generalizes across individual EEG patterns
- Adaptive advantage increases with fatigue (+9.0% to +11.2%) and holds across four fatigue model assumptions
- Half of patients habituate while half do not, validating the need for personalized, not fixed scheduling

Limitation: Real-data validation uses offline replay on recorded EEG, not live closed-loop streaming. The system makes decisions on real brain data but cannot observe the brain's response to those decisions.